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QUARTER DRONE BENCH FOR SYSTEM IDENTIFICATION AND CONTROL

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Abstract. ***The quarter drone or Aeropendulum system represents the dynamic of a quarter of a drone. The interest in this system is due to its nonlinearity and instability making a challenge to develop control strategies. Furthermore, due to the previously mentioned fact and that it is possible to use traditional control methods in at least one specific interval (0 degrees to 60 degrees) it is possible to use this system to teach introductory concepts related to both system identification and control theory, it is also possible to demonstrate more complex topics in both fields due to its nonlinearity and stability. Initially the physical structure of the aeropendulum was built, consisting of a base where two triangular towers hold up the pendulum's axis, connected to said axis is the pendulum's rod with the BLDC motor and counterweights. An Inertial Measurement Unit (IMU) sensor was installed at the axis as well as a multiturn potentiometer to measure the rotation of the system. After completing the physical structure and installing the sensors and motor, work began on the electronic system, where each component was connected to an Arduino Mega, in addition to installing the motor's power supply. Then, computational models were developed to run and simulate the system in software using MATLAB and Simulink. Finally, a Proportional Integral Derivative (PID) and a Pole Placement controllers were developed for the aforementioned region using data obtained by the system to estimate an adequate transfer function. Thus, it was possible to control the system in its more linear and stable regions in a satisfactory manner. In future work it is expected to also collect the IMU data in a meaningful manner in order to calculate the system's angular position with more precision and the development of more advanced control strategies.***

Keywords: *System Identification, Experimental Bench, Quarter Drone, Aeropendulum, Real Time Control.*

1. INTRODUCTION

The aeropendulum or quarter drone is an experimental bench that consists of a pendular rod that has an electric motor with a propeller attached to it. The motor produces thrust causing the pendulum to swing. As demonstrated by Regina *et al*. (2024) this system represents the dynamics of that of the quarter of a drone. This simplified structure, therefore, allows for testing and developing control strategies in a controlled environment.

Pendulums are inherently nonlinear systems, which makes it a specially interesting in the context of control theory, specifically the control of its position and stabilization in unstable regions (Rafiuddin, and Khan, 2022). Furthermore, as presented in Control Education for societal scale challenges: A community roadmap by Rossiter *et al* (2023), there is a

need for a more hands on approach to teaching control theory and its related fields. In this case the aeropendulum system can be used to teach both introductory and more advanced topics in both control theory and system identification.

One of the main objectives of this system is to analyze the behavior of a drone, by studying only one of its degrees of freedom. Therefore, it is important to understand the dynamic of the complete system. Such dynamics can be seen in the work of Foehn *et al* (2022) where the authors developed a drone to test different control strategies, by using high precision sensors integrating software and hardware to develop to produce efficient controllers.

On the other hand, Chen *et al* (2017) developed a similar system as that of the aeropendulum. An inverted pendulum, composed of two electric motors with two interconnected rods, having two degrees of freedom. The authors used MATLAB and Simulink to develop a system to control the inverted pendulum's angular position.

Furthermore, Regina *et al* (2024), in their work showcases that the aeropendulum, or quarter drone as the authors calls it represents the dynamic of one of the drone arms. By using a model developed in Solidworks, similar to the one proposed in this paper. The authors implement and compare two different control strategies based on their analysis of the quarter drone system and simulating the system by the combined use of MATLAB and Simulink, specifically the SIMSCAPE MULTIBODY toolbox.

Rossiter *et al* (2023) demonstrates the importance of hands-on learning for both control theory and its related fields. Due to the fact that control is a math heavy field and the traditional approach of starting by teaching mathematics instead of the application of control, many students can lose interest in the topic. In this work the use of pendular systems for students to control the angular position is highlighted, demonstrating that by learning with hands on experiments, it is possible to not only teach the fundamentals of control theory, but it is also possible to foster their interest in this subject.

Additionally, in another work Rossiter *et al* (2018) proposes a portable system with a dynamic inspired by a helicopter with three degrees of system. The purpose of the proposed system is as a challenge for students to control the angular position of the system. This system is composed of two rods, the first responsible for the yaw movement and rotation, stabilized by counterweights, while the second one with fans. Each fan requires a control system to manage the system's dynamic.

As demonstrated by Rossiter *et al* (2023), systems such as the one presented in this paper, is extremely important in both helping students learning control theory as well as fostering their interest. In this context the work presented shows the project involved in developing an aeropendulum, including its mechanical, electronic and software project. Consequently, the PID controllers and the estimated model can serve as a benchmark for other works or projects in this field.

This paper is organized in the following sections: Section 2 describes the methods employed for identifying the system, specifically the instrument variable method, and the control structure, PID and Pole Placement, used in this work. Section 3 contains the development of the aeropendulum system, divided into three different subsections: mechanical, electronic, and software, each going over how this system was developed, and their main components. Followed by section 4, which exposes the results obtained in identifying the system as well as the PID and Pole Placement controllers developed for the working region of 0° to 60°. Finally, section 5 concludes this paper and presents possible future work.

2. METHODS

2.1 EXPERIMENTAL MODELING OF LINEAR SYSTEM

To identify the system using its own data the function tfest from MATLAB's System identification toolbox was used. This function takes input and output data and uses it to estimate the transfer function's parameters using the Instrument Variable (IV) method. This method consists of using a correlation vector ζ(t), where each element in this vector is an instrument variable (Ljung, 1998). These elements are correlated to the regression variables and not to the noise. Ljung (1998) summarizes the model with the following set of equations: Eq.(1), Eq.(2), Eq.(3), and Eq.(4). Where ε_F is the prediction error, and $\hat{\theta}$ are the model parameters

$$\varepsilon_F(t,\theta) = L(q)[y(t) - \varphi^T(+)\theta] \tag{1}$$

$$\hat{\theta}_N^{IV} = sol_{\theta \in D_m}[f_N(\theta, Z^N) = 0] \tag{2}$$

Where:

$$f_N(\theta, Z^N) = \frac{1}{N}\sum_{T=1}^{N} \zeta(t,\theta)\alpha\big(\varepsilon_F(T,\theta)\big) \tag{3}$$

$$\zeta(t,\theta) = k_u(q,\theta)u(t) \tag{4}$$

2.2 PID STRUCTURE

According to Ogata (2014), a PID controller can be described in the following equation, Eq.(5), where K_P represents the proportional gain to the systems error, T_i the integral time, T_d the derivative time, $e(t)$ as the system error, and $u(t)$ as the control signal

$$u(t) = K_p e(t) + \frac{K_p}{T_i}\int_0^t e(t)dt + K_p T_d \frac{de(t)}{dt} \tag{5}$$

Using the Laplace transform it is possible to obtain a function that represents the ratio of the control signal by the system's error, this function only depends on the variable S from the Laplace transform, the proportional gain K_P, and both the derivative and integral time, as shown in Eq.(6).

$$\frac{U(s)}{E(s)} = K_p\left(1 + \frac{1}{T_i s} + T_d s\right) \tag{6}$$

Figure 1 below shows the diagram of a complete closed loop system with a controller. In this context $R(s)$ is the reference signal, $D(s)$: are any disturbances, and $C(s)$ is the system's output.

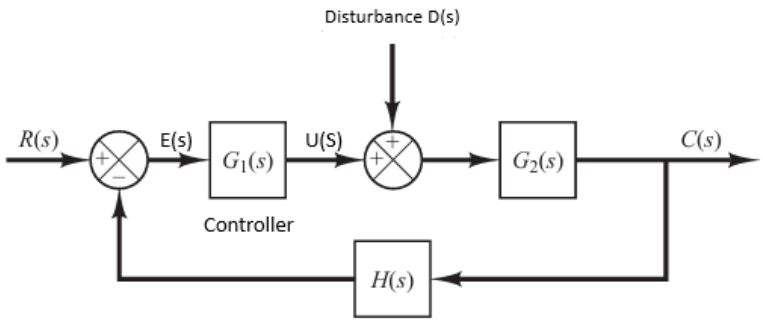

Figure 1. Control loop, from Ogata. (2014), Altered by the authors. (2025)

As seen in the previous figure, this system uses the feedback response to determine the error, the difference between the system's output and the applied reference, in order for the controller to minimize the difference between the reference and output. Furthermore, as demonstrated by Ogata (2014) the integrator, a pole placed at the systems origin, allows this system to reject step type disturbances. Additionally, the error in the steady state response is zero also due to the integral action (Ogata, 2014)

The PID structure was developed in MATLAB using the Control System Designer from the Control Systems Toolbox. Specifically, by using the RLTool user interface. RLtool allows the user to visualize the root locus of the system's transfer function. Therefore, allowing the design of a controller by changing the gain and placing zeros and poles to implement the PID structure as explained by Ogata (2014). This tool not only allows the user to insert design requirements to better help the method but also useful responses such as the system's step response and control signal.

2.3 POLE PLACEMENT

Pole placement is a control design method in which poles of the closed loop system is placed on the desired location through selecting an appropriate state feedback gain matrix (Ogata, 2014). The resulting closed loop system has no inputs, since its objective is to maintain the output at zero, resulting in a regulator system (Ogata, 2014). However, it is possible to insert an integrator in the feedforward path, between the error compensator and the plant in order for the system to track a reference (Ogata, 2014). The resulting control action can be calculated using the following equations, eq.(7), eq.(8):

$$\begin{bmatrix}\dot{x}\\ \dot{\xi}\end{bmatrix} = \begin{bmatrix}A - k_e C - BK_i & Bk_i\\ 0_{1\times n} & 0\end{bmatrix}\begin{bmatrix}\dot{x}\\ \xi\end{bmatrix} + \begin{bmatrix}0_{n\times 1} & K_e\\ 1 & -1\end{bmatrix}\begin{bmatrix}r\\ y\end{bmatrix} \tag{7}$$

$$[u] = \begin{bmatrix}-K & K_i\end{bmatrix}\begin{bmatrix}\hat{x}\\ \xi\end{bmatrix} + [0_{2\times 1}]\begin{bmatrix}r\\ y\end{bmatrix} \tag{8}$$

Where, $([\hat{x}\ \xi]^T)$ are the system's states and $([r\ y]^T)$ are the system's inputs. Furthermore, it is important to note that it is necessary to check the system's stability by checking the eigenvalue of the resulting gain matrix (Ogata, 2014). Finally, it is possible to transform the resulting system as transfer functions (Ogata, 2014).

3. PROJECT

The development of this system can be divided into three subprojects: mechanical, electronics, and software. In the mechanical project the physical structure of the aeropendulum was developed, including the base for its axis as well as the structures for fixing both the sensors and the motors, and finally the counterweights. In the electronic project each sensor and motor were tested and connected to the microcontroller, additionally the power supply used was selected. Finally, by using MATLAB and Simulink, code to send input signals and save the data produced by the sensors was developed.

3.1 MECHANICAL PROJECT

The system was built on top of a compressed wood board, serving as its base. Two triangular towers made of Medium Density Fiberboard (MDF), where the side faces of each tower are rectangular triangles, their back face are scalene triangles. Each face was glued together at their edges, a 3D printed structures were also attached to the corners of each tower, those structures were then screwed into the base fixing in place both towers.

In order to attach the aluminum axis to both towers, a rectangular cuboid was designed to have a ball bearing, thus allowing the axis to move freely as well as a fixture to attach a multiturn potentiometer to the axis to capture its rotation. In the middle of the axis a hole was drilled and threaded to allow the aeropendulum's rod to be screwed into place. Furthermore, a 3D printed housing for the Inertial Measurement Unit (IMU) was developed and attached to the axis.

The threaded rod also allowed a relatively easy way to attach and remove counterweights, each counterweight was drilled and threaded to easily fit into the rod. A 3D printed support was also made to attach the motor in the aeropendulum's rod. For the motor to produce thrust a fixed pitch two bladed propellers were attached to the motor using an O-ring. Figure 2 shows both the CAD model and the finished structure.

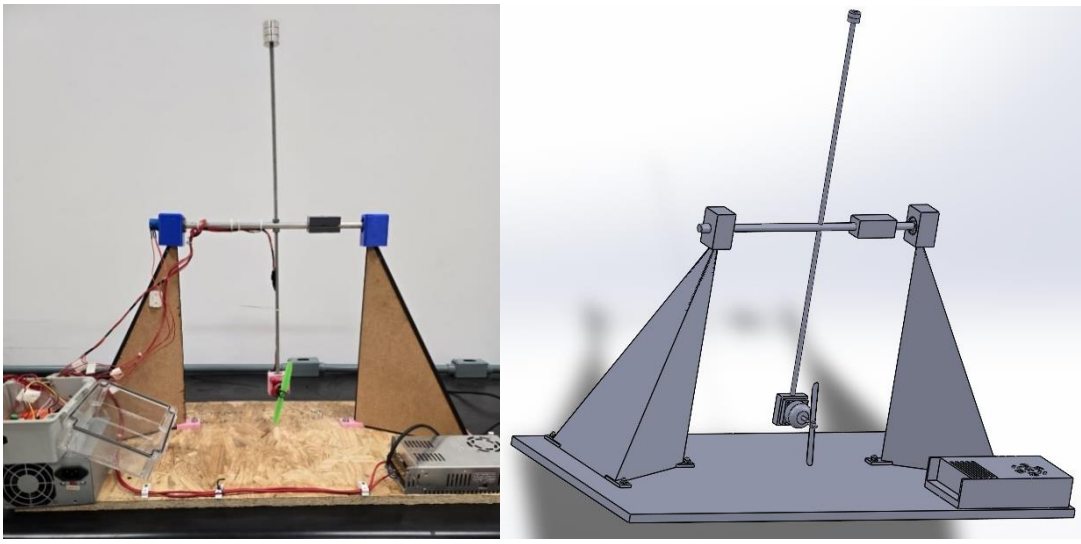

Figure 2. finished structure (left) and CAD model (right), from the Authors. (2025)

3.2 ELECTRONIC PROJECT

After the physical structure of the aeropendulum was complete the main sensors were installed in their respective housings previously designed, as well as attaching the BLDC motor and its Electronic Speed Controller (ESC). Afterwards, a Switch Mode Power Supply (SMPS) used to power the motor was fixed into the base of the system. Afterwards, 1mm wires were used to solder the main components into a DB 15M VGA, and 2mm wires with banana plugs were used to wire the SMPS. Table 1 below showcases each component used and their models.

Table 1. Components, Source: The authors. (2025)

Components	**Model**
Microcontroller	Arduino Mega 2560
IMU	ICM 20948
Motor	Brushless DC Motor CF2805
Motor driver	8-bit ESC (Electronic Speed Controller)
Potentiometer	10 kΩ Multiturn potentiometer
Potentiometer	10 kΩ Potentiometer
Power Supply	12 V 50 A SMPS (Switch Mode Power Supply)
Wires	1mm and 2mm wires
Connectors	DB 15M VGA Banana jacks and plugs Screw terminal
Communication cable	USB-A to USB-B cable

Subsequently, with all the components installed a control box containing an Arduino Mega 2560 with a USB-A to USB-B cable to properly connect with a computer, the female connector of the DB 15M VGA, and banana jacks to properly connect the power supply. Furthermore, a Printed Circuit Board (PCB) was made in order to organize the incoming signals from the VGA connecter and into the Arduino Mega, the following Fig. 3 showcases the PCB developed.

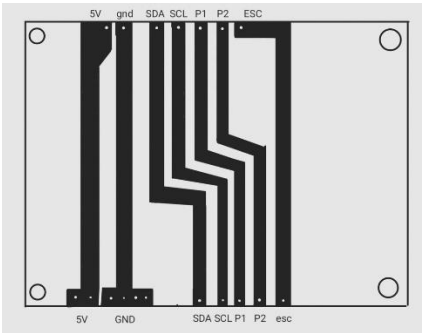

Figure 3. Printed circuit board, from the Authors. (2025)

Finally, the connections from the PCB were connected to the Arduino, the BLDC signal was connected in the digital Pin 4, the multiturn potentiometer was connected to the analog Pin 2, the manual control potentiometer into analog Pin 3, the IMU's SCL and SDA pins were connected in pins 21/SCL and 20/SDA respectively. Whilst the potentiometer was connected to Arduino’s 5V supply the imu was connected in the 3.3V supply. Finally, all the components ground was connected in the same ground pin. Figure 4 below shows the schematic of the electronic system.

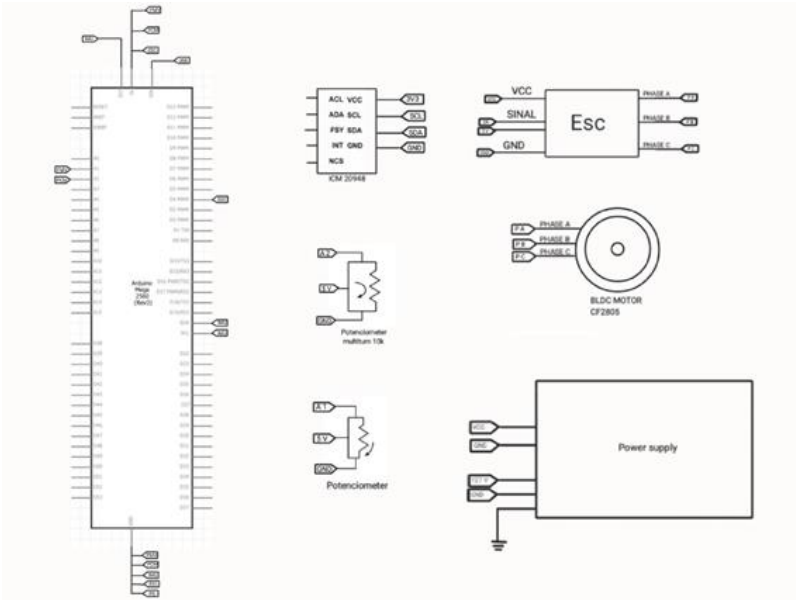

Figure 4. Electronic schematic diagram, from the Authors. (2025)

3.3 SOFTWARE PROJECT

In the software project, three different Simulink models were developed, initially an open loop model to perform tests and identify the plant. Following the identification a closed loop simulation model was developed to test and validate the controllers. Finally, a closed loop model was developed for the system using the angle calculated from the multiturn to close the loop and consequently control the system. Beyond these Simulink models, a MATLAB script was developed to configure the systems parameters such as calibration values, filter, run time, and sampling frequency. The run time and sampling frequency default values were defined to be 60 seconds and 20 Hz respectively. Both the open loop model and closed loop model that runs in the hardware uses the following subsystem showcased in Fig. 5 below to allow for an easier way to interact with the system.

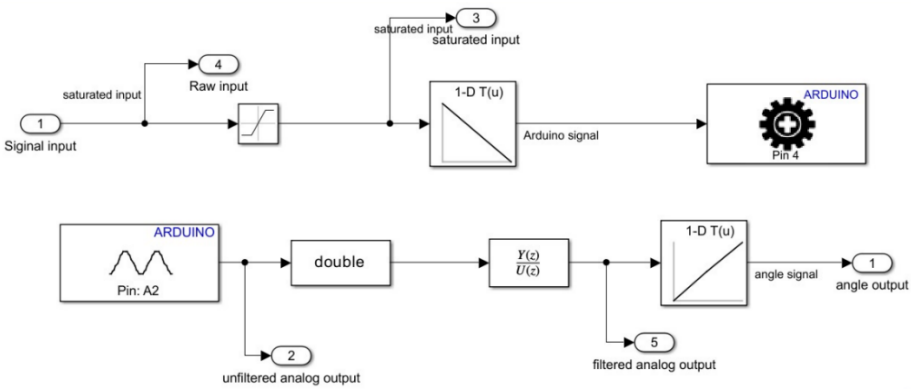

Figure 5. Aeropendulum Subsystem, from the Authors. (2025)

To read the angular position of the pendulum's rod, the subsystem uses the values produced by the multiturn potentiometer installed in the axis of the system. Since the Arduino's ADC is 10 bits the values produced are in a range of 0 to 1023, additionally the reference for the ADC is set to 1.1V to capture subtle changes in the angle. Since this data has some noise, both inherently caused by the potentiometer's jittering as well as receiving some of the noise caused by the BLDC motor it was necessary to develop a simple first order Butterworth filter implemented in the system as a discrete transfer function. Finally, the data is converted to an angular value in degrees by a 1D-Look Up Table (1D-LUT) using the calibration parameters. Generally, calibration is done by measuring the analog values in the following angles: 0°, 90°, and 180°. As a reference 0° degrees is considered when the system is at rest with the motor at the bottom.

On the other hand, the Aeropendulum subsystem accepts signals that range from negative one to positive one as the BLDC's input. In this context, the negative one is the motor at maximum velocity clockwise, whilst positive one is the

motor at maximum velocity anticlockwise, whilst zero represents the motor at 0% velocity or on idle. This signal is then converted into a range of 0° to 180°, since ESCs and Servos use the same type of signal to communicate with the microcontroller, a servo write block is used to send the signals to the ESC, thus setting the speed of the motor. Since the motor at 0.3, or 30% of its maximum velocity is more than enough for the aeropendulum system to do a full loop in the counterclockwise direction and -0.2 or -20% is also enough to loop in the clockwise direction the signal is saturated to those values, especially because any value greater than them could damage the system.

Both models, the open loop model and closed loop model, in other words, the models that use the physical system run in the microcontroller itself. Through the option "Monitor & Tune" in Simulink it is possible to compile the Simulink models to a target hardware and monitor input and output signals using serial communication. Thus, allowing the real time visualization of the input and output signals as well as the possibility to save the signals in the workspace using a bus and the to workspace block.

4. RESULTS

4.1 MODELING

To develop a model of the Aeropendulum system a step input was used. The initial value of the step was 0%, while the final value was 12% of the motor's speed, the step happened at 30 seconds. As shown in Fig. 6 below, using this input the system stabilizes around 45 degrees.

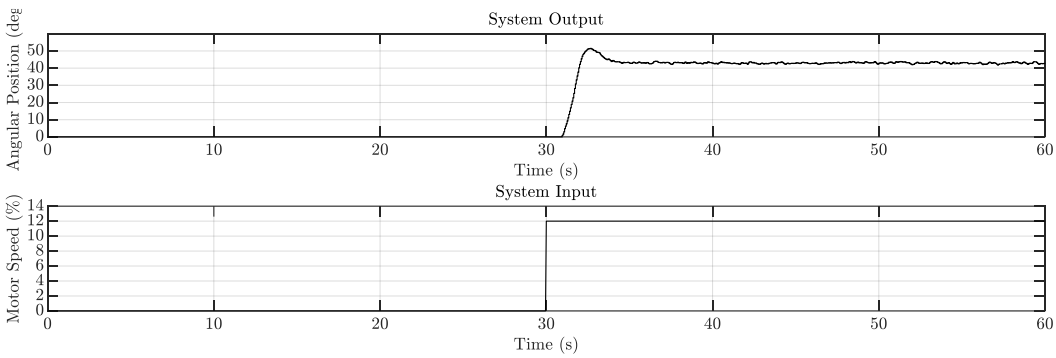

Figure 6. step response in the 0 to 45 degrees region, from the Authors. (2025)

Additionally, another acquisition was made for the same region. The main difference being that a small step of 5% of the motor's speed was used to "warm-up" the motor, allowing it to start spinning whilst remaining in place. Afterwards, another step was applied, increasing the motor's speed to 11%, and consequently allowing the system angular position to move near 45 degrees. Figure 7 below showcases the second step and the resulting angular position the first step and its resulting angular position was subtracted to separate it from the warm-up step.

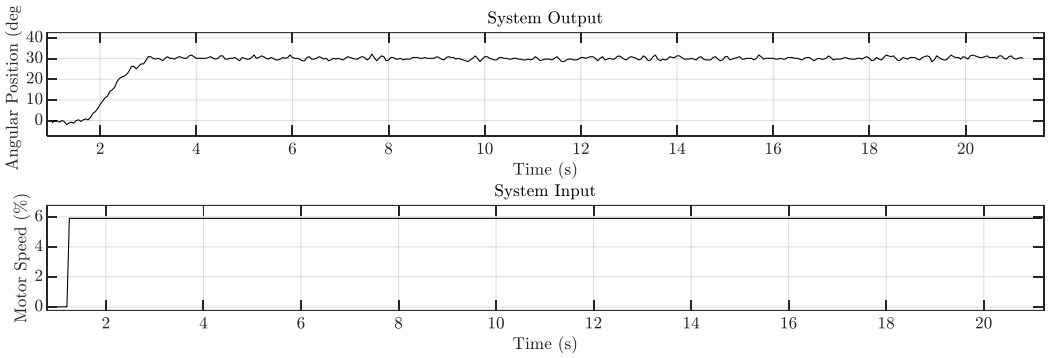

Figure 7. step response with warm-up, from the Authors. (2025)

By using the data obtained in the first test it is possible to identify the system in the 0 to 45 degrees region, using MATLAB's tfest function to estimate continuous transfer functions that fit the dynamic of the system. Multiple transfer functions were estimated, varying the order between one and ten, and including the dead time observed. Without the dead time, transfer functions with orders greater than seven had a negative fit so they were removed, with the 7th order transfer function having the greatest fit of 97.64%. On the other hand, when including dead time, it is important to note

that transfer functions with orders greater than two had a negative fit as well, consequently they were discarded, while the 2nd order transfer function had the best fit of 95.41%. Figure 8 below showcases the dynamic of the fourth order transfer function and the fit of the transfer functions without dead time.

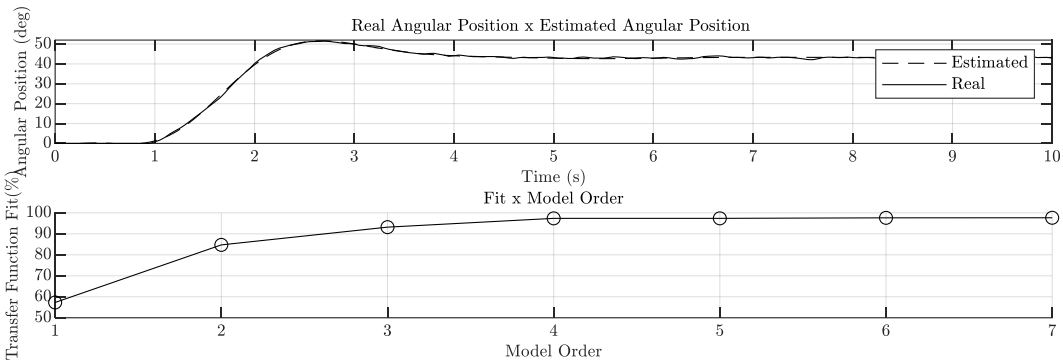

Figure 8. Estimated transfer function response for the 0 to 45 degrees region, from the Authors. (2025)

Repeating the same procedure for the second dataset, transfer functions were estimated, varying their orders between one and ten, as well as including the dead time observed. Similarly to the other experiment, transfer functions with orders greater than 7 were discarded due to their negative fit. Without dead time the best fit was 90.41% of the 7th order transfer function. When including dead time, transfer functions with orders above third order were discarded to the negative fit. The best fit obtained with dead time was 89.68%, of the second order transfer function Fig. 9 below showcases the response of the second order transfer function as well as the respective fits of the transfer functions with dead time.

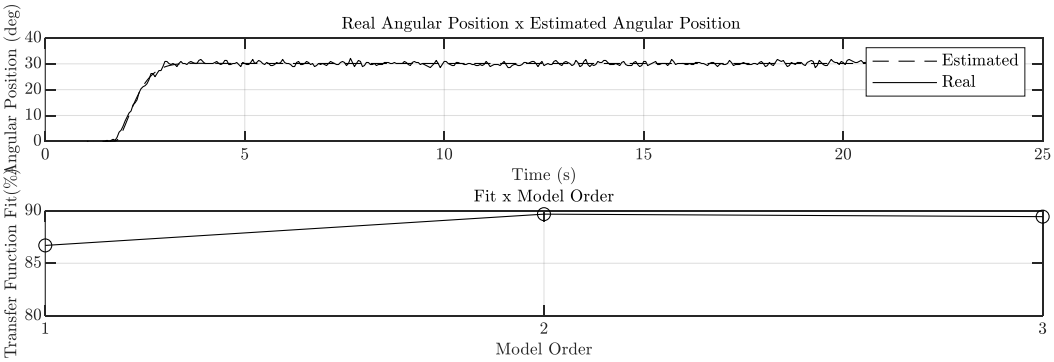

Figure 9. Estimated transfer function response for the second dataset, from the Authors. (2025)

4.2 CONTROL

To design the controller the following requirements were defined:

1) The control signal must be in the range of -0.3 to 0.3.
2) Setting time of at most 20 seconds.
3) Maximum overshoot of 5%.

Using the 4th order transfer function of the first dataset previously modeled, a Proportional Integral Derivative (PID) controller, and a pole placement controller were developed. The reference signal that they needed to follow was a step sequence in which the steps had the following amplitudes: 60°, 30°, -60°, -30°. To meet the design requirements, first for PID controller, MATLAB's Control System Designer (CSD) tool was used, specifically their Root Locus (RL) tools that allow the visualization of the system's root locus as well as the system's response and the control signal. It's important to mention that for this controller, the transfer function used for simulation was designed without time delay. By editing the controller, placing a pole at the origin as well as two zeros in an empirical manner in the real axis, and changing the gain to satisfy the design requirements.

The following Fig. 10 shows the result of the PID controller without any disturbances present, both the simulated result using the transfer function used to develop said controller as well as the actual systems response with the controller. As well as the control signal and error for both the simulation and the aeropendulum itself.

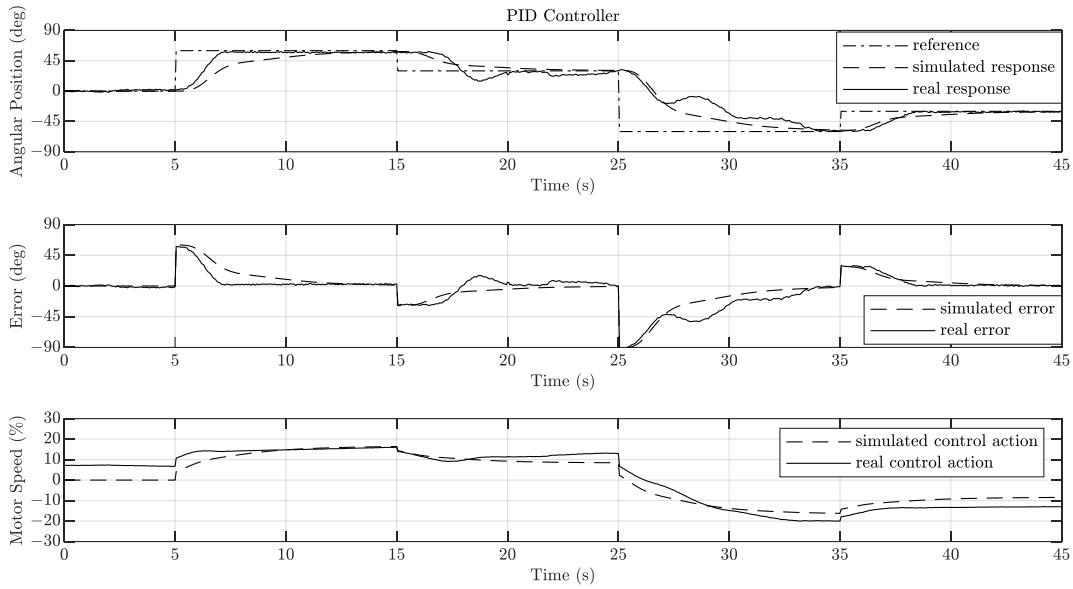

Figure 10. PID controller angular position, Error and Control signal, from the Authors. (2025)

Using the second order transfer function with dead time from the second dataset, a pole placement controller was developed using a Matlab script, with the same design requirements as the PID. The pole placement controller was simulated and tested on the aeropendulum. Figure 11 below shows the simulated response as well as the actual response from the system.

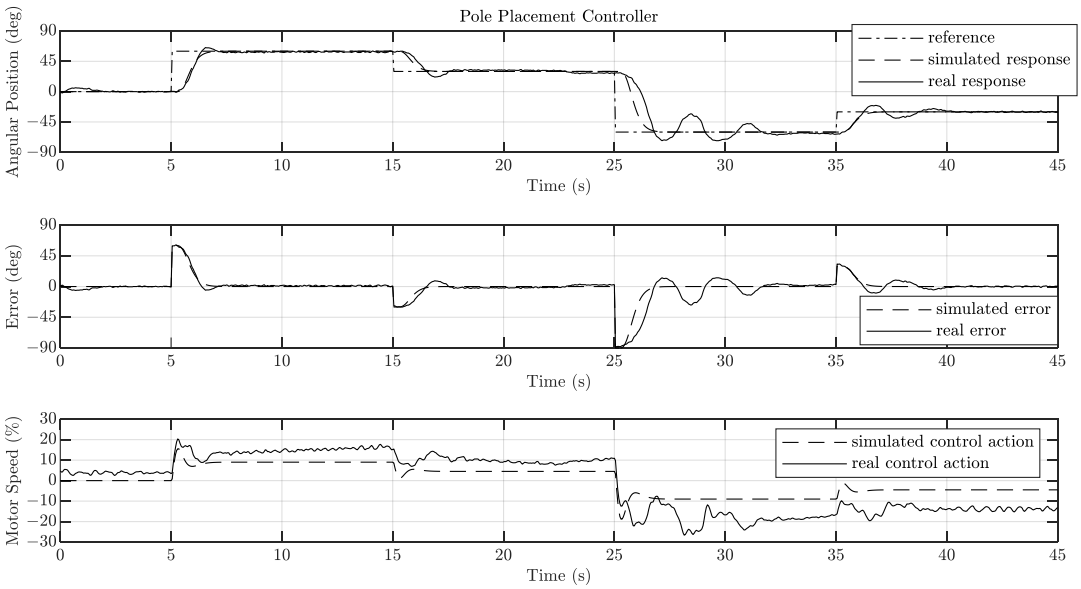

Figure 11. Pole placement controller angular position, Error and Control signal, from the Authors. (2025)

It is important to note that the pole placement controller had a better performance than the PID controller. As seen in Fig. 11, and Fig 10. The pole placement controller stabilized faster than the PID controller after the first step, it is specially visible in the negative region, where the PID controller barelly reaches the reference, negative 60 degrees. Furthermore, the pole placement controller had a similar response in both the simulation and the actual physical system.

5. CONCLUSION

In conclusion, this paper shows that the aeropendulum or quarter drone system can be easily built, calibrated and used in real time. By performing system identification using the data collected from the system it is also possible to develop a model to simulate its response in the 60 to -60 degrees region. Consequently, allowing the design of two controllers,

one PID and another pole placement controller, that work both in simulation and in the actual system. Furthermore, the response of the pole placement controller was considerably superior to that of the PID.

Therefore, due to these reasons this system can be used to demonstrate and teach in a practical manner introductory concepts of both system identification and control theory, furthermore since it is a nonlinear unstable system it can also help demonstrate more complex and advanced topics of both fields.

In the future, it is expected to implement more advanced control methods such as adaptive control through a gain scheduler. Furthermore, it is expected to use the IMU sensor already installed in the aeropendulum to calculate the angle of rod with a greater accuracy and precision, adapting the Simulink models to use said sensors instead of the multiturn potentiometer.

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